# FR306x Datasheet

Highly Secure Dual Mode Bluetooth 5.3 MCU with CAN FD

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FREQCHIP

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# DESCRIPTION

The FR306x product family is a low-power , highly secure, single-chip wireless MCU that integrates a high performance BR/EDR/BLE radio and CAN FD for Automotive and Industrial applications. It complies with the AEC-Q100-Grade2 automotive standard.

FR306x compliant with Bluetooth Specification V5.3 support BR 1Mbps GFSK, EDR 2Mbps π/4-DQPSK, 3Mbps 8DPSK; BLE 1M/2Mbps GFSK, 125K/500K mode. Support independent opening and closing different mode, support Bluetooth multi-role and multi-link.

FR306x support AUTOSAR platform. FR306x has up to 4 independent CAN FD controllers built in and is compatible with CAN2.0 A / B.

# FEATURES

- Bluetooth Specification 5.3
  - RX sensitivity -93dBm (BR 1Mbps)
  - RX sensitivity -94dBm (EDR 2Mbps)
  - RX sensitivity -85dBm (EDR3Mbps)
  - RX sensitivity -97dBm (LE 1Mbps)
  - RX sensitivity -93dBm (LE 2Mbps)
  - RX sensitivity -100.5dBm (LE S2)
  - RX sensitivity -102.5dBm (LE S8)
  - TX Power -20dBm ~ 10dBm
- Dedicated Dual-Core processor architecture
  - MC1(CM33) core with FPU & DSP instruction, Max frequency is 156MHz, as the main processor
  - 48MHz 32-bit RISC core for BT+BLE stack with stand alone SRAM
- Embedded 512KB SRAM for CM33 only
- XIP Nor Flash up to 2MB for user application, with 32KB cache
- QSPI/OPI interface with max 78M serial clock
- SoC Peripherals
  - Up to 4-ch PDM interface for Digital Audio
  - 3x I2C
  - 5x UART with flow controller
  - 2x I2S ports
  - 2x SPI Master support QSPI,OSPI
  - 2x SPI Slave
  - RGB565/888
  - SDIO3.0/eMMC 4.5.1
  - USB-OTG
  - Up to 31 x PWM
  - Up to 9-ch SAR-ADC (FR3068E-C/FR3068EP-C/FR3068E-D is 9 channels, while FR3066DQ-C/FR3066E-D is 7 9channels)
  - PWM Digital Audio Modulation Unit (PSD_DAC): supports PWM stereo differential output
  - Configurable GPIO up to 56
- CAN FD Controller
  - Conform with CAN protocol version 2.0 part A, B and ISO 11898-1
  - CAN FD with up to 64 data bytes
  - AUTOSAR support
  - SAE J1939 support
- Hardware Encryption engine
  - AES-128, AES-192, AES-256
- Low-power consumption (DCDC buck 3.3 V, 25°C)
  - MC1(CM33)Typical active core current: < 9 mA at 96MHz(<50uA/MHz)
  - Transceiver current (Buck mode, 3.3 V supply)
    - Typical RX: 9.8 mA
    - Typical TX at 0 dBm: 18.2 mA

- In Sleep mode, RTC is enabled, and any GPIO can wake up: 8.9 uA@64KB SRAM retention, 7.2uA @ 32KB SRAM retention
- Average power consumption (BLE) for 1-second broadcast interval: 28.8uA (transmission power 0 dBm)
- Average power consumption of 1-second connection (BLE):20.5uA (transmission power 0 dBm)

# TYPICAL APPLICATIONS

- Automotive electronics
  - — Secure Car Access
  - — Keyless Entry
  - — Wireless Battery Management Systems (WBMS)
- Industry
  - — Positioning/Localization
  - — Building Control and Monitoring
  - — Process/Factory Automation
  - — Access Control

# ORDERING INFORMATION

| Order Code | MC1(CM33) frequency | Flash ROM | SRAM | GPIO | CAN | Temperature Range | Package | Transport Media,Quantity | MSL | AEC-Q100 |
|---|---|---|---|---|---|---|---|---|---|---|
| FR3066D-C | 156MHz | 1MB | 256KB | 29 | 2 | -40°C ~ +105°C | QFN48 6*6 | Tape and Reel, 5000 | MSL3 | N |
| FR3066DQ-C | 156MHz | 1MB | 256KB | 29 | 2 | -40°C ~ +105°C | QFN48 6*6 | Tape and Reel, 5000 | MSL3 | Y |
| FR3066E-D | 156MHz | 2MB | 512KB | 29 | 3 | -40°C ~ +105°C | QFN48 6*6 | Tape and Reel, 5000 | MSL3 | N |
| FR3068E-C | 156MHz | 2MB | 512KB | 56 | 2 | -40°C ~ +105°C | QFN80 9*9 | Tape and Reel, 3000 | MSL3 | N |
| FR3068EP-C | 156MHz | 2MB | 512KB | 56 | 2 | -40°C ~ +85°C | QFN80 9*9 | Tape and Reel, 3000 | MSL3 | N |
| FR3068E-D | 156MHz | 2MB | 512KB | 56 | 4 | -40°C ~ +105°C | QFN80 9*9 | Tape and Reel, 3000 | MSL3 | N |
| FR3068EQ-D | 156MHz | 2MB | 512KB | 53 | 4 | -40°C ~ +105°C | QFN80 9*9 | Tape and Reel, 3000 | MSL3 | Y |

# 1. Hardware Details

## 1.1 Block Diagram

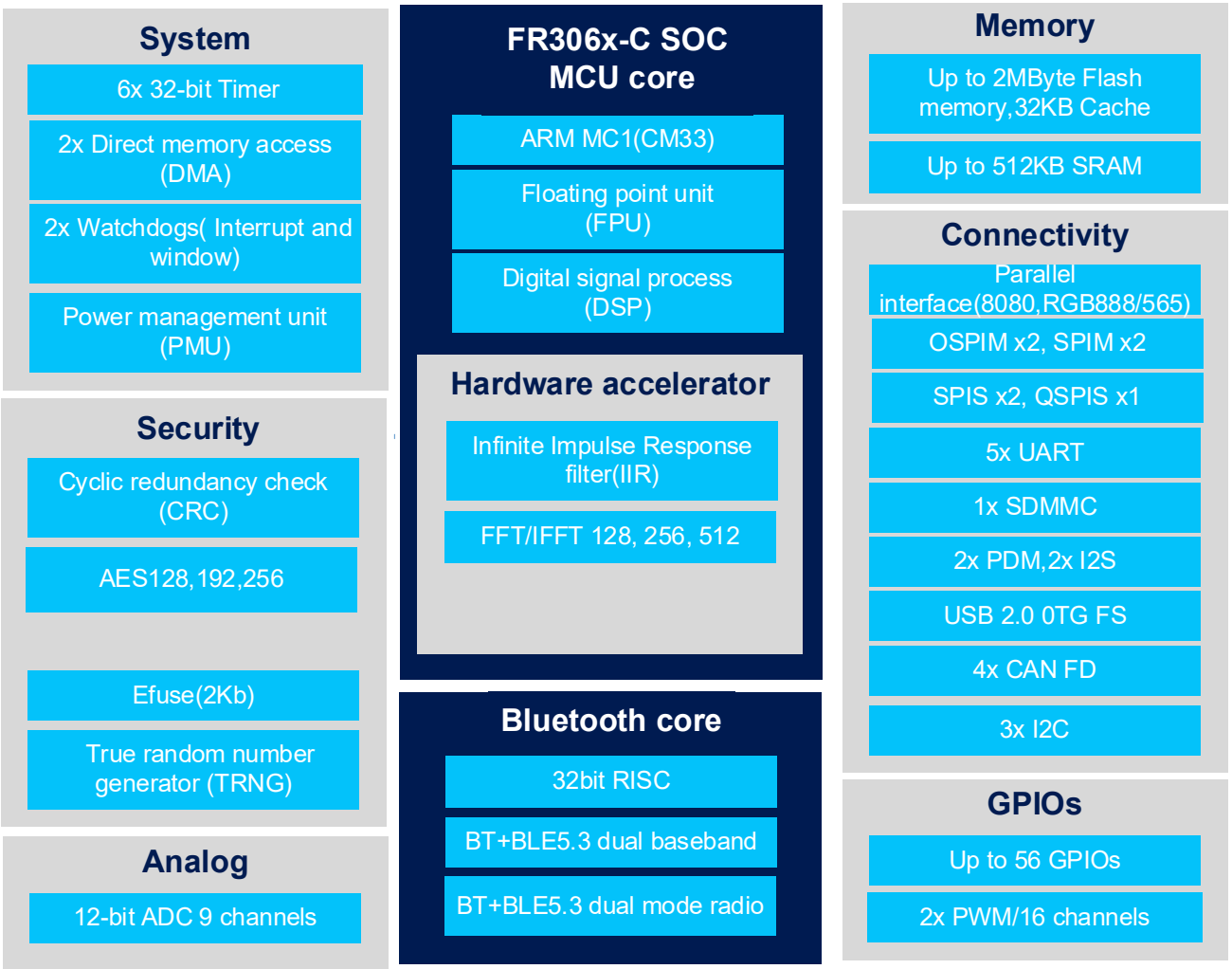



Figure 1-1 Block Diagram

## 1.2 Hardware Resources

| Model / Resource | FR3066D-C/ FR3066DQ-C | FR3066E-D | FR3068E-C | FR3068E-D/ FR3068EQ-D | FR3068EP-C |
|---|---|---|---|---|---|
| Package | QFN48 | QFN48 | QFN80 | QFN80 | QFN80 |
| Frequency | 156MHz | 156MHz | 156MHz | 156MHz | 156MHz |
| GPIO | 29 | 29 | 56 | 56(FR3068E-D)<br>53(FR3068EQ-D) | 56 |
| FLASH | 1MB | 1MB | 2MB | 2MB | 2MB |
| SRAM | 256KB | 512KB | 512KB | 512KB | 512KB |
| PSRAM | 0 | 0 | 0 | 0 | 2MB |
| General-purpose Timer | 6 | 6 | 6 | 6 | 6 |
| PWM | 27 | 27 | 31 | 31 | 31 |
| CAN FD | 2 | 3 | 2 | 4 | 2 |
| UART | 5 | 5 | 5 | 5 | 5 |
| I2C | 3 | 3 | 3 | 3 | 3 |
| SPIM | 2 | 2 | 2 | 2 | 2 |
| SPIS | 2 | 2 | 2 | 2 | 2 |
| I2S | 2 | 2 | 2 | 2 | 2 |
| OSPI(XIP) | 0 | 0 | 1 | 1 | 1 |
| QSPI(XIP) | 0 | 0 | 1 | 1 | 1 |
| PDM | 2 channels | 2 channels | 4 channels | 4 channels | 4 channels |
| PWM audio output | 2 channels | 2 channels | 2 channels | 2 channels | 2 channels |
| USB 2.0 FS | 1 | 1 | 1 | 1 | 1 |
| SDIO | 1 | 1 | 1 | 1 | 1 |
| 12-bit ADC | 8 channels | 8 channels | 9 channels | 9 channels | 9 channels |

## 1.3 Bluetooth RF Transceiver

- Qualified to Dual Mode Bluetooth V5.3 specification
- -93dBm (BR 1M) receive sensitivity
- -94dBm (EDR 2M) receive sensitivity
- -85dBm (EDR 3M) receive sensitivity
- -97dBm (LE 1M) receive sensitivity
- -93dBm (LE 2M) receive sensitivity
- -100.5dBm (LE S2) receive sensitivity

- -102.5dBm (LE S8) receive sensitivity
- TX Power -20dBm ~ 10dBm
- Integrated channel filters
- Digital demodulator for improved sensitivity and co-channel rejection
- Real time digitized RSSI
- Fast AGC for enhanced dynamic range

## 1.4 Bluetooth Controller

- All device classes support (Broadcaster, Central, Observer, Peripheral)
- All packet types(Advertising/Data/Control)
- Encryption(AES/CCM)
- Bit stream processing (CRC, Whitening)
- Frequency hopping calculation
- Supports power down of the baseband during the protocol's idle periods

## 1.5 CAN Communication Controller

- Conform with CAN protocol version 2.0 part A, B and ISO 11898-1
- CAN FD with up to 64 data bytes supported
- AUTOSAR support
- SAE J1939 support
- Improved acceptance filtering
- Two configurable Receive FIFOs
- Up to 64 dedicated Receive Buffers
- Up to 32 dedicated Transmit Buffers
- Configurable Transmit FIFO
- Configurable Transmit Queue
- Configurable Transmit Event FIFO

## 1.6 Physical Interfaces

- UART port for Debugging and AT Commands
- 1/2/4-bit SPI flash memory interface to support external SPI flash
- I2C interface to support external EEPROM or other devices (like FM Receiver)
- QSPI interface with max 78MHz serial clock
- Up to 56 general purpose IOs
- SDIO/eMMC (4.5) Flash interface supports either 1/4/8-bit mode

- General purpose 9-ch 12-bit ADC used for AD-Key and other analog input
- PWM controller
- General purpose programmable timer for various task
- Watchdog used for tracking unexpected exception

## 1.7 Integrated Power Control and Regulation

- Embedded Power-On-Reset
- Low power core voltage
- On-chip high efficiency switch-mode power supply, 2.9v to 3.6v input direct from battery and programmable output voltage
- On-chip Low Dropout (LDO) Linear Regulator for internal Digital, RF and Analog circuit.
- Power management features include software shutdown and hardware wake-up
- Power-on-reset cell detects low supply voltage

## 1.8 Package

- QFN80 9*9
- QFN48 6*6

# 2. Hardware Package Information

## 2.1 Package

### 2.1.1 FR3066D-C /FR3066DQ-C / FR3066E-D

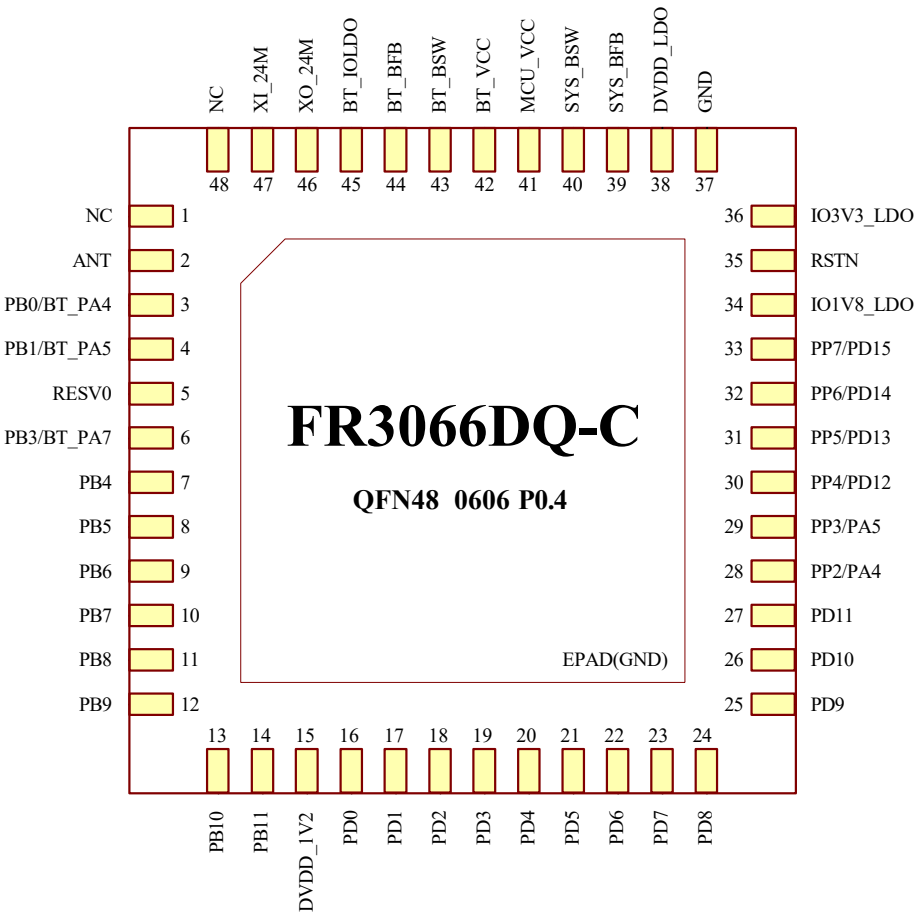



Figure 2-1 FR3066D-C /FR3066DQ-C / FR3066E-D Pin Information

### 2.1.2 FR3068E-C / FR3068EP-C / FR3068E-D

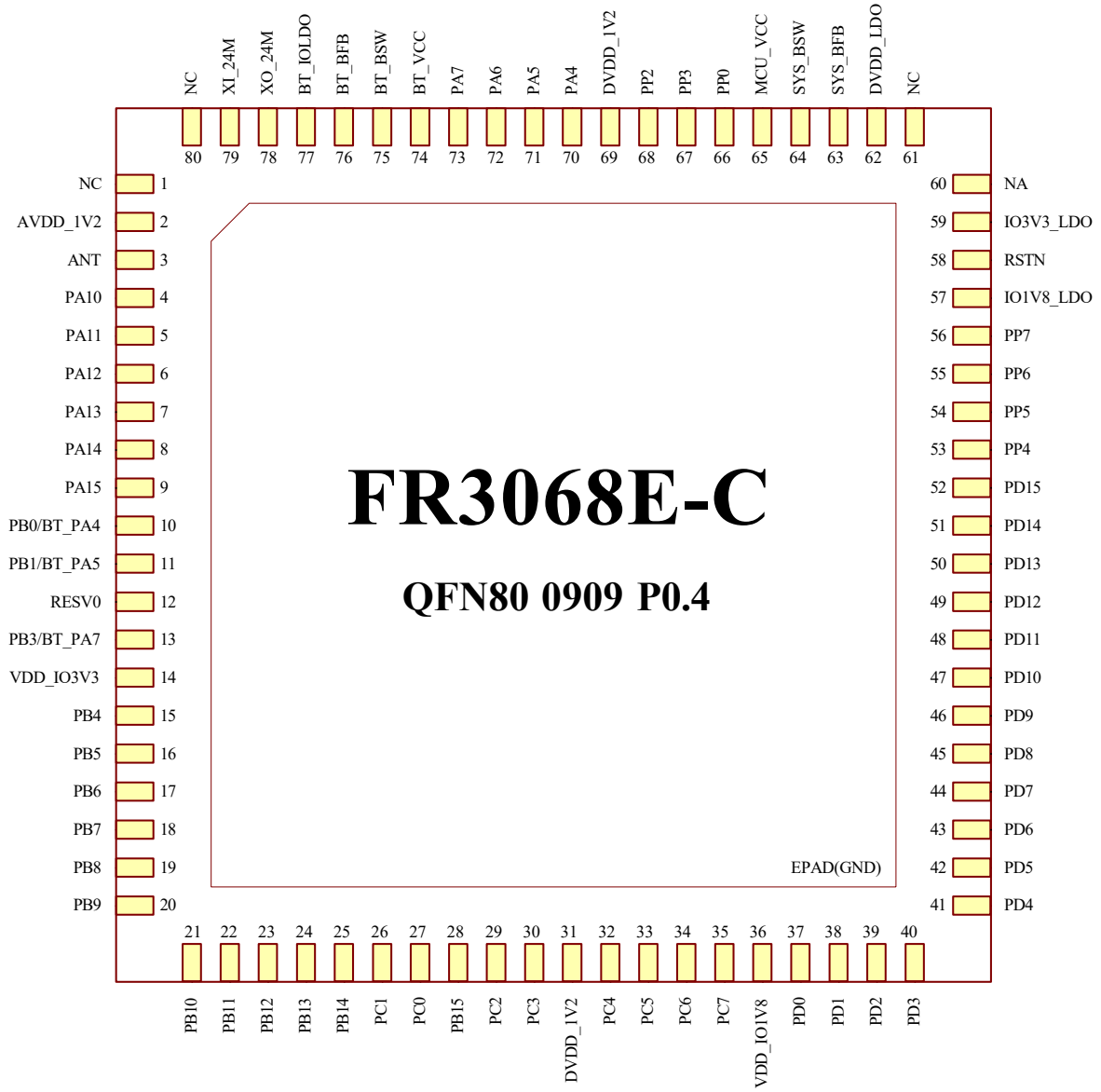



Figure 2-2 FR3068E-C / FR3068EP-C/ FR3068E-D Pin Information

### 2.1.3 FR3068EQ-D

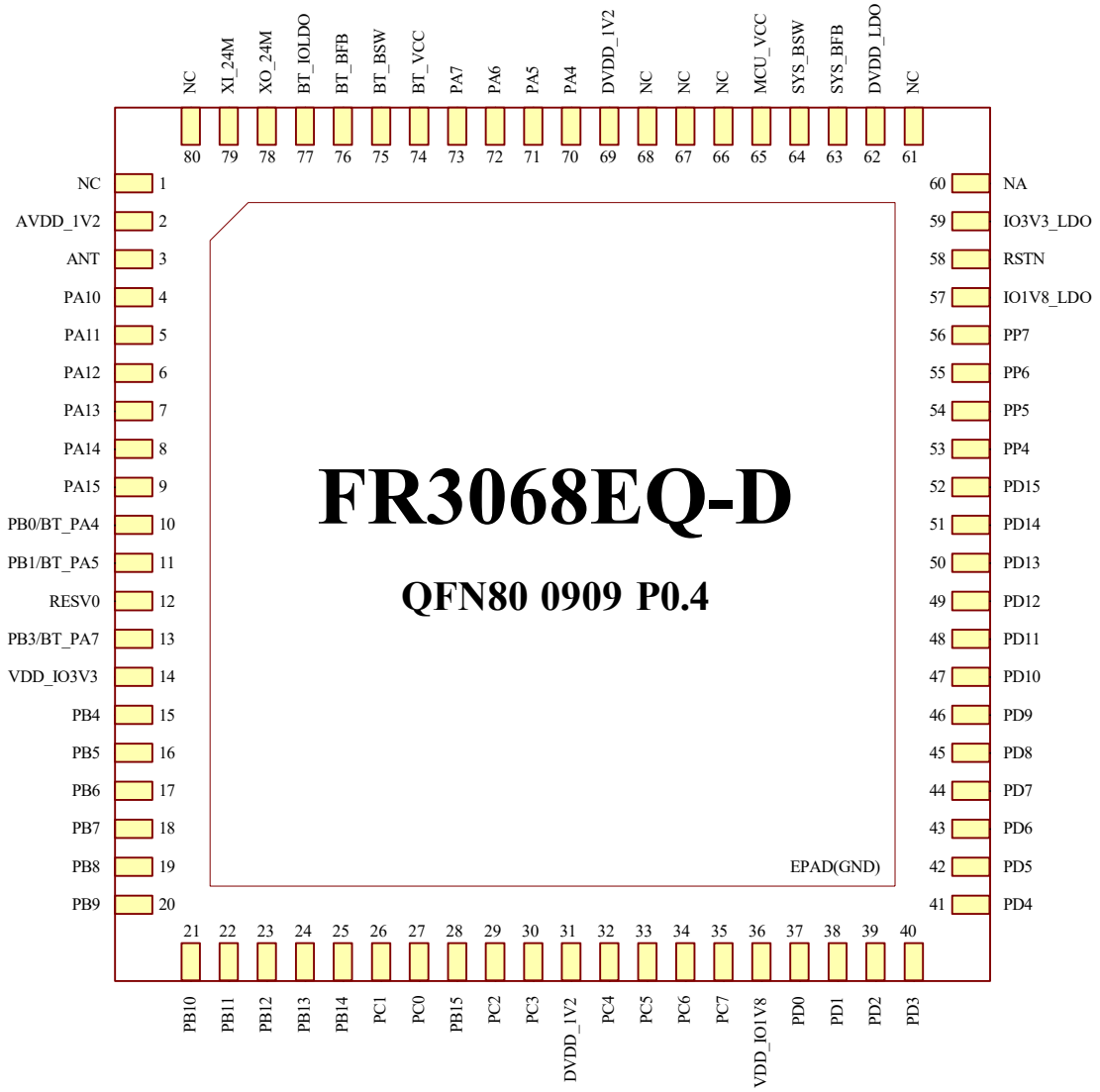
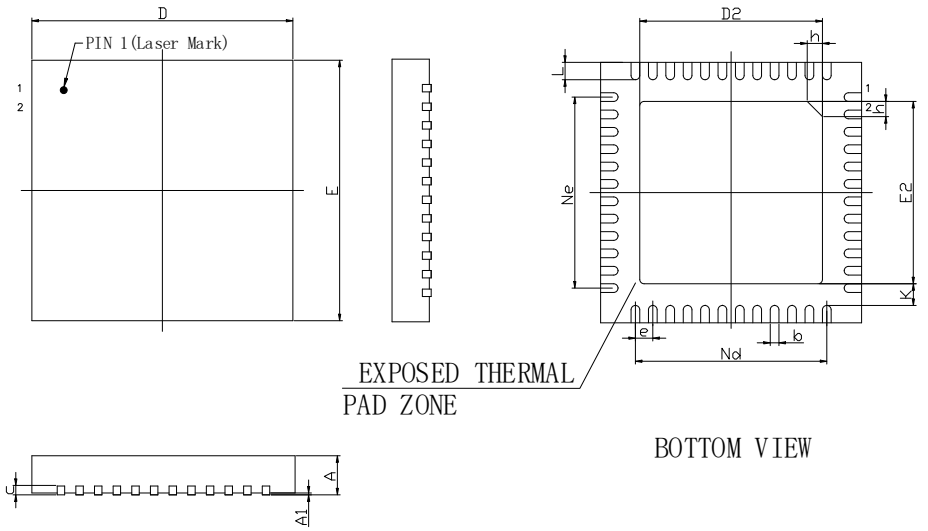



Figure 2-3 FR3068EQ-D

## 2.2 Package Physical Dimensions

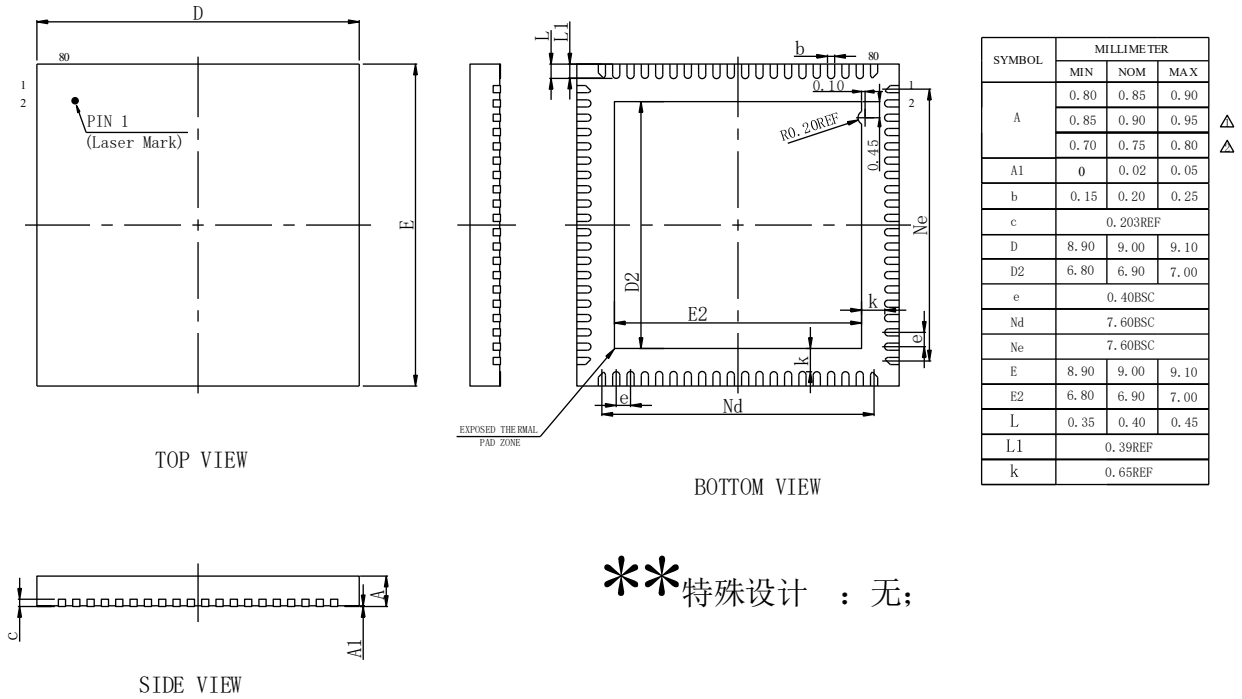
### 2.2.1 FR3066D-C /FR3066DQ-C / FR3066E-D




| SYMBOL | MILLIMETER | | |
|---|---|---|---|
| | MIN | NOM | MAX |
| A | 0.85 | 0.90 | 0.95 |
| A1 | 0 | 0.02 | 0.05 |
| b | 0.15 | 0.20 | 0.25 |
| c | 0.18 | 0.20 | 0.23 |
| D | 5.90 | 6.00 | 6.10 |
| D2 | 4.10 | 4.20 | 4.30 |
| e | 0.40BSC | | |
| Ne | 4.40BSC | | |
| Nd | 4.40BSC | | |
| E | 5.90 | 6.00 | 6.10 |
| E2 | 4.10 | 4.20 | 4.30 |
| L | 0.35 | 0.40 | 0.45 |
| K | 0.20 | 0.50 | 0.55 |
| h | 0.30 | 0.35 | 0.40 |
| L/F载体尺寸 (MIL) | 177*177 | | |

Figure 2-4 FR3066D-C /FR3066DQ-C / FR3066E-D Package Physical Dimensions

### 2.2.2 FR3068E-C / FR3068EP-C / FR3068E-D / FR3068E-D




| SYMBOL | MILLIMETER | | |
|---|---|---|---|
| | MIN | NOM | MAX |
| A | 0.80 | 0.85 | 0.90 |
| | 0.85 | 0.90 | 0.95 △ |
| | 0.70 | 0.75 | 0.80 △ |
| A1 | 0 | 0.02 | 0.05 |
| b | 0.15 | 0.20 | 0.25 |
| c | 0.203REF | | |
| D | 8.90 | 9.00 | 9.10 |
| D2 | 6.80 | 6.90 | 7.00 |
| e | 0.40BSC | | |
| Nd | 7.60BSC | | |
| Ne | 7.60BSC | | |
| E | 8.90 | 9.00 | 9.10 |
| E2 | 6.80 | 6.90 | 7.00 |
| L | 0.35 | 0.40 | 0.45 |
| L1 | 0.39REF | | |
| k | 0.65REF | | |

**特殊设计 ：无；

Note: FR3068E-C thickness 0.90mm

Figure 2-5 FR3068E-C / FR3068EP-C / FR3068E-D / FR3068EQ-D Package Physical Dimensions

## 2.3 Pins Description

Table 2-1 Pin Abbreviations

| Notation | Description |
|---|---|
| AI | Analog input |
| AO | Analog output |
| IO | Bidirectional(digital) |
| PWR | Power |
| GND | Ground |

### 2.3.1 FR3066D-C /FR3066DQ-C / FR3066E-D Pin Description

Table 2-2 FR3066D-C /FR3066DQ-C / FR3066E-D Pin Description

| Pin Number | Pin Name | Type | Description |
|---|---|---|---|
| 1 | NC | - | Not connected |
| 2 | ANT | Analog | Bluetooth antenna |
| 3 | PB0/BT_PA4 | IO | Multi-Function GPIO |
| 4 | PB1/BT_PA5 | IO | Multi-Function GPIO |
| 5 | RESV0 | - | Reserved pin0 |
| 6 | PB3/BT_PA7 | IO | Multi-Function GPIO |
| 7 | PB4 | IO | Multi-Function GPIO |
| 8 | PB5 | IO | Multi-Function GPIO |
| 9 | PB6 | IO | Multi-Function GPIO |
| 10 | PB7 | IO | Multi-Function GPIO |
| 11 | PB8 | IO | Multi-Function GPIO |
| 12 | PB9 | IO | Multi-Function GPIO |
| 13 | PB10 | IO | Multi-Function GPIO |
| 14 | PB11 | IO | Multi-Function GPIO |
| 15 | DVDD_1V2 | Analog | Digital 1.2V power supply |
| 16 | PD0 | IO | Multi-Function GPIO |
| 17 | PD1 | IO | Multi-Function GPIO |
| 18 | PD2 | IO | Multi-Function GPIO |
| 19 | PD3 | IO | Multi-Function GPIO |
| 20 | PD4 | IO | Multi-Function GPIO |

| Pin Number | Pin Name | Type | Description |
|---|---|---|---|
| 21 | PD5 | IO | Multi-Function GPIO |
| 22 | PD6 | IO | Multi-Function GPIO |
| 23 | PD7 | IO | Multi-Function GPIO |
| 24 | PD8 | IO | Multi-Function GPIO |
| 25 | PD9 | IO | Multi-Function GPIO |
| 26 | PD10 | IO | Multi-Function GPIO |
| 27 | PD11 | IO | Multi-Function GPIO |
| 28 | PP2/PA4 | IO | Multi-Function GPIO |
| 29 | PP3/PA5 | IO | Multi-Function GPIO |
| 30 | PP4/PD12 | IO | Multi-Function GPIO |
| 31 | PP5/PD13 | IO | Multi-Function GPIO |
| 32 | PP6/PD14 | IO | Multi-Function GPIO |
| 33 | PP7/PD15 | IO | Multi-Function GPIO |
| 34 | IO1V8_LDO | Analog | 1.8V IO LDO |
| 35 | RSTN | Analog | Chip reset pin, low active（built-in pull-up resistor 10K） |
| 36 | IO3V3_LDO | Analog | 3.3V IO LDO |
| 37 | GND | GND | Ground |
| 38 | DVDD_LDO | Analog | Digital 1.2V LDO output |
| 39 | SYS_BFB | Analog | System DC/DC feedback input terminal |
| 40 | SYS_BSW | Analog | System DC/DC output terminal |
| 41 | MCU_VCC | PWR | MCU power supply |
| 42 | BT_VCC | PWR | BT power supply |
| 43 | BT_BSW | Analog | BT DC/DC output terminal |
| 44 | BT_BFB | Analog | BT DC/DC feedback input terminal |
| 45 | BT_IOLDO | Analog | BT IOLDO |
| 46 | XO_24M | Analog | 24M Crystal oscillator output |
| 47 | XI_24M | Analog | 24M Crystal oscillator input |
| 48 | NC | - | Not connected |

### 2.3.2 FR3068E-C / FR3068EP-C / FR3608E-D Pin Description

Table 2-3 FR3068E-C / FR3608E-D Pin Description

| Pin Number | Pin Name | Type | Description |
|---|---|---|---|
| 1 | NC | - | Not connected |
| 2 | AVDD_1V2 | Analog | Analog 1.2V power supply |
| 3 | ANT | Analog | Bluetooth antenna |
| 4 | PA10 | IO | Multi-Function GPIO |
| 5 | PA11 | IO | Multi-Function GPIO |
| 6 | PA12 | IO | Multi-Function GPIO |
| 7 | PA13 | IO | Multi-Function GPIO |
| 8 | PA14 | IO | Multi-Function GPIO |
| 9 | PA15 | IO | Multi-Function GPIO |
| 10 | PB0/BT_PA4 | IO | Multi-Function GPIO |
| 11 | PB1/BT_PA5 | IO | Multi-Function GPIO |
| 12 | RESV0 | - | Reserved pin0 |
| 13 | PB3/BT_PA7 | IO | Multi-Function GPIO |
| 14 | VDD_IO3V3 | Analog | GPIO 3.3V power supply |
| 15 | PB4 | IO | Multi-Function GPIO |
| 16 | PB5 | IO | Multi-Function GPIO |
| 17 | PB6 | IO | Multi-Function GPIO |
| 18 | PB7 | IO | Multi-Function GPIO |
| 19 | PB8 | IO | Multi-Function GPIO |
| 20 | PB9 | IO | Multi-Function GPIO |
| 21 | PB10 | IO | Multi-Function GPIO |
| 22 | PB11 | IO | Multi-Function GPIO |
| 23 | PB12 | IO | Multi-Function GPIO |
| 24 | PB13 | IO | Multi-Function GPIO |
| 25 | PB14 | IO | Multi-Function GPIO |
| 26 | PC1 | IO | Multi-Function GPIO |
| 27 | PC0 | IO | Multi-Function GPIO |
| 28 | PB15 | IO | Multi-Function GPIO |
| 29 | PC2 | IO | Multi-Function GPIO |
| 30 | PC3 | IO | Multi-Function GPIO |
| 31 | DVDD_1V2 | Analog | Digital 1.2V power supply |

| Pin Number | Pin Name | Type | Description |
|---|---|---|---|
| 32 | PC4 | IO | Multi-Function GPIO |
| 33 | PC5 | IO | Multi-Function GPIO |
| 34 | PC6 | IO | Multi-Function GPIO |
| 35 | PC7 | IO | Multi-Function GPIO |
| 36 | VDD_IO1V8 | Analog | 1.8V IO power supply |
| 37 | PD0 | IO | Multi-Function GPIO |
| 38 | PD1 | IO | Multi-Function GPIO |
| 39 | PD2 | IO | Multi-Function GPIO |
| 40 | PD3 | IO | Multi-Function GPIO |
| 41 | PD4 | IO | Multi-Function GPIO |
| 42 | PD5 | IO | Multi-Function GPIO |
| 43 | PD6 | IO | Multi-Function GPIO |
| 44 | PD7 | IO | Multi-Function GPIO |
| 45 | PD8 | IO | Multi-Function GPIO |
| 46 | PD9 | IO | Multi-Function GPIO |
| 47 | PD10 | IO | Multi-Function GPIO |
| 48 | PD11 | IO | Multi-Function GPIO |
| 49 | PD12 | IO | Multi-Function GPIO |
| 50 | PD13 | IO | Multi-Function GPIO |
| 51 | PD14 | IO | Multi-Function GPIO |
| 52 | PD15 | IO | Multi-Function GPIO |
| 53 | PP4 | IO | Multi-Function GPIO |
| 54 | PP5 | IO | Multi-Function GPIO |
| 55 | PP6 | IO | Multi-Function GPIO |
| 56 | PP7 | IO | Multi-Function GPIO |
| 57 | IO1V8_LDO | Analog | 1.8V IO LDO |
| 58 | RSTN | Analog | Chip reset pin, low active（built-in pull-up resistor 10K） |
| 59 | IO3V3_LDO | Analog | 3.3V IO LDO |
| 60 | NA | - | Not used |
| 61 | NC | - | Not connected |
| 62 | DVDD_LDO | Analog | Digital 1.2V LDO output |
| 63 | SYS_BFB | Analog | System DC/DC feedback input terminal |
| 64 | SYS_BSW | Analog | System DC/DC output terminal |
| 65 | MCU_VCC | PWR | MCU power supply |
| 66 | PP0 | IO | Multi-Function GPIO |

| Pin Number | Pin Name | Type | Description |
|---|---|---|---|
| 67 | PP3 | IO | Multi-Function GPIO |
| 68 | PP2 | IO | Multi-Function GPIO |
| 69 | DVDD_1V2 | Analog | Digital 1.2V power supply |
| 70 | PA4 | IO | Multi-Function GPIO |
| 71 | PA5 | IO | Multi-Function GPIO |
| 72 | PA6 | IO | Multi-Function GPIO |
| 73 | PA7 | IO | Multi-Function GPIO |
| 74 | BT_VCC | PWR | BT power supply |
| 75 | BT_BSW | Analog | BT DC/DC output terminal |
| 76 | BT_BFB | Analog | BT DC/DC feedback input terminal |
| 77 | BT_IOLDO | Analog | BT IOLDO |
| 78 | XO_24M | Analog | 24M Crystal oscillator output |
| 79 | XI_24M | Analog | 24M Crystal oscillator input |
| 80 | NC | - | Not connected |

### 2.3.3 FR3608EQ-D Pin Description

| Pin Number | Pin Name | Type | Description |
|---|---|---|---|
| 1 | NC | - | Not connected |
| 2 | AVDD_1V2 | Analog | Analog 1.2V power supply |
| 3 | ANT | Analog | Bluetooth antenna |
| 4 | PA10 | IO | Multi-Function GPIO |
| 5 | PA11 | IO | Multi-Function GPIO |
| 6 | PA12 | IO | Multi-Function GPIO |
| 7 | PA13 | IO | Multi-Function GPIO |
| 8 | PA14 | IO | Multi-Function GPIO |
| 9 | PA15 | IO | Multi-Function GPIO |
| 10 | PB0/BT_PA4 | IO | Multi-Function GPIO |
| 11 | PB1/BT_PA5 | IO | Multi-Function GPIO |
| 12 | RESV0 | - | Reserved pin0 |
| 13 | PB3/BT_PA7 | IO | Multi-Function GPIO |
| 14 | VDD_IO3V3 | Analog | GPIO 3.3V power supply |
| 15 | PB4 | IO | Multi-Function GPIO |

| Pin Number | Pin Name | Type | Description |
|---|---|---|---|
| 16 | PB5 | IO | Multi-Function GPIO |
| 17 | PB6 | IO | Multi-Function GPIO |
| 18 | PB7 | IO | Multi-Function GPIO |
| 19 | PB8 | IO | Multi-Function GPIO |
| 20 | PB9 | IO | Multi-Function GPIO |
| 21 | PB10 | IO | Multi-Function GPIO |
| 22 | PB11 | IO | Multi-Function GPIO |
| 23 | PB12 | IO | Multi-Function GPIO |
| 24 | PB13 | IO | Multi-Function GPIO |
| 25 | PB14 | IO | Multi-Function GPIO |
| 26 | PC1 | IO | Multi-Function GPIO |
| 27 | PC0 | IO | Multi-Function GPIO |
| 28 | PB15 | IO | Multi-Function GPIO |
| 29 | PC2 | IO | Multi-Function GPIO |
| 30 | PC3 | IO | Multi-Function GPIO |
| 31 | DVDD_1V2 | Analog | Digital 1.2V power supply |
| 32 | PC4 | IO | Multi-Function GPIO |
| 33 | PC5 | IO | Multi-Function GPIO |
| 34 | PC6 | IO | Multi-Function GPIO |
| 35 | PC7 | IO | Multi-Function GPIO |
| 36 | VDD_IO1V8 | Analog | 1.8V IO power supply |
| 37 | PD0 | IO | Multi-Function GPIO |
| 38 | PD1 | IO | Multi-Function GPIO |
| 39 | PD2 | IO | Multi-Function GPIO |
| 40 | PD3 | IO | Multi-Function GPIO |
| 41 | PD4 | IO | Multi-Function GPIO |
| 42 | PD5 | IO | Multi-Function GPIO |
| 43 | PD6 | IO | Multi-Function GPIO |
| 44 | PD7 | IO | Multi-Function GPIO |
| 45 | PD8 | IO | Multi-Function GPIO |
| 46 | PD9 | IO | Multi-Function GPIO |
| 47 | PD10 | IO | Multi-Function GPIO |
| 48 | PD11 | IO | Multi-Function GPIO |
| 49 | PD12 | IO | Multi-Function GPIO |
| 50 | PD13 | IO | Multi-Function GPIO |

| Pin Number | Pin Name | Type | Description |
|---|---|---|---|
| 51 | PD14 | IO | Multi-Function GPIO |
| 52 | PD15 | IO | Multi-Function GPIO |
| 53 | PP4 | IO | Multi-Function GPIO |
| 54 | PP5 | IO | Multi-Function GPIO |
| 55 | PP6 | IO | Multi-Function GPIO |
| 56 | PP7 | IO | Multi-Function GPIO |
| 57 | IO1V8_LDO | Analog | 1.8V IO LDO |
| 58 | RSTN | Analog | Chip reset pin, low active（built-in pull-up resistor 10K） |
| 59 | IO3V3_LDO | Analog | 3.3V IO LDO |
| 60 | NA | - | Not used |
| 61 | NC | - | Not connected |
| 62 | DVDD_LDO | Analog | Digital 1.2V LDO output |
| 63 | SYS_BFB | Analog | System DC/DC feedback input terminal |
| 64 | SYS_BSW | Analog | System DC/DC output terminal |
| 65 | MCU_VCC | PWR | MCU power supply |
| 66 | NC | - | Not connected |
| 67 | NC | - | Not connected |
| 68 | NC | - | Not connected |
| 69 | DVDD_1V2 | Analog | Digital 1.2V power supply |
| 70 | PA4 | IO | Multi-Function GPIO |
| 71 | PA5 | IO | Multi-Function GPIO |
| 72 | PA6 | IO | Multi-Function GPIO |
| 73 | PA7 | IO | Multi-Function GPIO |
| 74 | BT_VCC | PWR | BT power supply |
| 75 | BT_BSW | Analog | BT DC/DC output terminal |
| 76 | BT_BFB | Analog | BT DC/DC feedback input terminal |
| 77 | BT_IOLDO | Analog | BT IOLDO |
| 78 | XO_24M | Analog | 24M Crystal oscillator output |
| 79 | XI_24M | Analog | 24M Crystal oscillator input |
| 80 | NC | - | Not connected |

# 3. RF Characteristics

Table 3-1 RF Performance Scope

| Item | Sym. | Min. | Typ. | Max. | Unit | Condition |
|---|---|---|---|---|---|---|
| Output Signal Cross-over Voltage | - | 2400 | - | 2483.5 | MHz | Programmable in 1 MHz |
| Data rate | BR 1Mbps<br>EDR 2Mbps<br>EDR 3Mbps<br>LE 1Mbps<br>LE 2Mbps<br>LE 125Kbps<br>LE 500Kbps | | | | | |

Table 3-2 RF Performance Characteristics

| Data Rate | Item | Sym. | Min. | Typ. | Max. | Unit | Condition |
|---|---|---|---|---|---|---|---|
| BR RF_Rx performance | Sensitivity | | | -93 | | dBm | - |
| | Maximum received signal<br>At 0.1%BER | | | 0 | | dBm | |
| | Co-channel C/I | | | 8 | | dB | Wanted signal at -60dBm |
| | Adjacent channel C/I<br>Image = –2MHz | +1/-1 MHz offset | | -5/-5 | | dB | |
| | | +2 MHz offset | | -38 | | dB | |
| | | +3/-3 MHz offset | | -41/-40 | | dB | Wanted signal at -67dBm |
| | Image rejection | -2 MHz offset | | -34 | | dB | |
| BR RF_TX performance | Output power, maximum setting | | -20 | 0 | 10 | dBm | |
| | Transmitter power control step | | 2 | | 8 | dB | |
| | Output power, control range | | | 30 | | dB | |
| | Initial carrier frequency offset | | | ±5 | | kHz | |
| | Frequency drift | | | ±10 | | kHz | |

| Data Rate | Item | Sym. | Min. | Typ. | Max. | Unit | Condition |
|---|---|---|---|---|---|---|---|
| | Frequency Deviation | Δf1avg | | 160 | | kHz | 140 kHz ~ 175 kHz |
| | | Δf2max | 115 | | | kHz | ⩾ 115 kHz |
| | | Δf2avg/Δf1avg | 0.8 | | | | ≥ 0.8 |
| | Adjacent channel power(ACP) | IM -NI = 2 | | -50 | | dBm | |
| | | IM -NI = 3 | | -54 | | dBm | |
| EDR RF_RX performance | Sensitivity | EDR2 2Mbps | | -94 | | dBm | |
| | | EDR3 3Mbps | | -85 | | dBm | |
| | Maximum received signal at 0.1% BER | | | 0 | | dBm | |
| EDR2 | Co-channel C/I | | | 15 | | dB | Wanted signal at -60dBm |
| | Adjacent channel C/I Image = –2MHz | +1/-1 MHz offset | | -7/-8 | | dB | |
| | | +2 MHz offset | | -30 | | dB | |
| | | +3/-3 MHz offset | | -40/-40 | | dB | Wanted signal at -67dBm |
| | Image rejection | -2 MHz offset | | -22 | | dB | |
| EDR3 | Co-channel C/I | | | 18 | | dB | Wanted signal at -60dBm |
| | Adjacent channel C/I Image = –2MHz | +1/-1 MHz offset | | 0/-1 | | dB | |
| | | +2 MHz offset | | -25 | | dB | |
| | | +3/-3 MHz offset | | -33/-33 | | dB | Wanted signal at -67dBm |
| | Image rejection | -2 MHz offset | | -17 | | dB | |
| EDR RF_TX performance | Output power, maximum setting | | | 7 | | dBm | |
| | Transmitter power control step | | 2 | | 8 | dB | |

| Data Rate | Item | Sym. | Min. | Typ. | Max. | Unit | Condition |
|---|---|---|---|---|---|---|---|
| | Output power, control range | | | 27 | | dB | |
| | Initial carrier frequency offset | | | ±5 | | kHz | |
| | Frequency drift | EDR2(2DH5) | | ±10 | | kHz | |
| | | EDR3(3DH5) | | ±10 | | kHz | |
| | EDR2 Adjacent channel power (ACP) | IM -NI = 2 | | -28 | | dBm | |
| | | IM -NI = 3 | | -40 | | | |
| | EDR3 Adjacent channel power (ACP) | IM -NI = 2 | | -28 | | dBm | |
| | | IM -NI = 3 | | -40 | | | |
| | EDR2 modulation accuracy | RMS DEVM | | 6 | | % | ≤20% |
| | | PEAK DEVM | | 13 | | | ≤30% |
| | | 99% DEVM | | 10 | | | ≤35% |
| | EDR3 modulation accuracy | RMS DEVM | | 6 | | % | ≤13% |
| | | PEAK DEVM | | 14 | | | ≤25% |
| | | 99% DEVM | | 10 | | | ≤20% |
| BLE 1Mbps RF_RX Performance (±250kHz deviation) | Sensitivity | 1 Mbps | | -97 | | dBm | |
| | Co-channel C/I | | | 6 | | dB | Wanted signal at -67dBm |
| | Adjacent channel C/I Image = –2MHz | +1/-1 MHz offset | | -6/-5 | | dB | |
| | | +2 MHz offset | | -42 | | dB | |
| | | >=3 MHz offset | | -46 | | dB | |
| | Image rejection | -2 MHz offset | | -28 | | dB | |
| BLE 1Mbps RF_TX Performance | Output power, maximum setting | | | 10 | | dBm | |
| | Output power, minimum setting | | | -20 | | dBm | |
| | Programmable output power range | | | 30 | | dB | |

| Data Rate | Item | Sym. | Min. | Typ. | Max. | Unit | Condition |
|---|---|---|---|---|---|---|---|
| | Modulation 20dB bandwidth | | | 1.2 | | MHz | |
| BLE 2Mbps RF_RX Performance (±500kHz deviation | Sensitivity | | | -93 | | dBm | |
| | Co-channel C/I | | | 7 | | dB | Wanted signal at -67dBm |
| | Adjacent channel C/I Image = –4MHz | +2/-2 MHz Offset | | -6/-5 | | dB | |
| | | +4 MHz Offset | | -30 | | dB | |
| | | >=6 MHz Offset | | -33 | | dB | |
| | Image rejection | -4 MHz Offset | | -21 | | dB | |
| BLE 2Mbps RF_TX Performance | Output power, maximum setting | | | 10 | | dBm | |
| | Output power, minimum setting | | | -20 | | dBm | |
| | Programmable output power range | | | 30 | | dB | |
| | Modulation 20dB Bandwidth | | | 2.3 | | MHz | |
| BLE 500 kbps RF_RX Performance (±250kHz deviation) | Sensitivity | | | -100.5 | | dB | |
| | Co-channel C/I | | | 7 | | dB | Wanted signal at -72dBm |
| | Adjacent channel C/I Image = –2MHz | +1/-1 MHz offset | | -8/-8 | | dB | |
| | | +2 MHz offset | | -43 | | dB | |
| | | >=3 MHz offset | | -46 | | dB | |
| | Image rejection | -2 MHz offset | | -40 | | dB | |
| BLE 500 kbps RF_TX Performance | Output power, maximum setting | | | 10 | | dBm | |

| Data Rate | Item | Sym. | Min. | Typ. | Max. | Unit | Condition |
|---|---|---|---|---|---|---|---|
| | Output power, minimum setting | | | -20 | | dBm | |
| | Programmable output power range | | | 30 | | dB | |
| | Modulation 20dB Bandwidth | | | 1.2 | | MHz | |
| BLE 125 kbps RF_RX Performance (±250kHz deviation) | Sensitivity | | | -102.5 | | dBm | |
| | Co-channel C/I | | | 1 | | dB | Wanted signal at -79dBm |
| | Adjacent channel C/I Image = –2MHz | +1/-1 MHz offset | | -9/-10 | | dB | |
| | | +2 MHz offset | | -48 | | dB | |
| | | >=3 MHz offset | | -55 | | dB | |
| | Image rejection | -2 MHz offset | | -40 | | dB | |
| BLE 125 kbps RF_TX Performance | Output power, maximum setting | | | 10 | | dBm | |
| | Output power, minimum setting | | | -20 | | dBm | |
| | Programmable output power range | | | 30 | | dB | |
| | Modulation 20dB Bandwidth | | | 1.2 | | MHz | |

# 4. Electrical Characteristics

## 4.1 Power on/off Operating Conditions

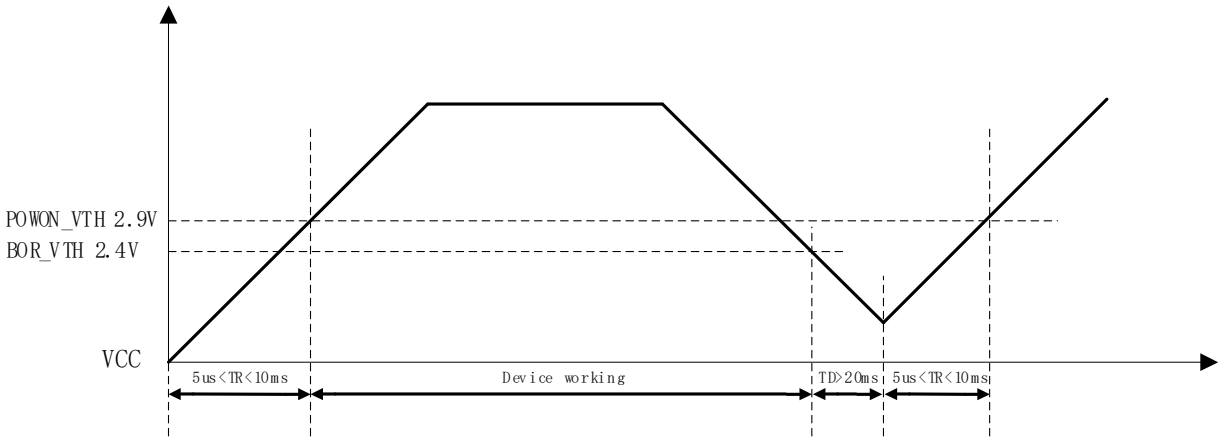



Figure 4-1 Power On/Off Timing Diagram

table 4-1 Power on/off Operating Conditions

| Parameter | Description and Conditions |
|---|---|
| VCC | Power supply |
| POWON_VTH | Power-on startup voltage |
| BOR_VTH | Power-off reset voltage |
| TR | Power on time. Require greater than 5us and less than 10ms |
| TD | After the VCC voltage drops below BOR_VTH, it is necessary to wait for TD to exceed 20ms for effective internal reset of the system. |

## 4.2 Recommended Operating Conditions

Table 4-2 Recommended Operating Conditions

| **Operating Condition** | | **Min.** | **Typ.** | **Max.** | **Unit** |
|---|---|---|---|---|---|
| Operating Temperature Range | | -40 | 25 | 105<br>85(FR3068EP-C) | ℃ |
| Output Voltage | IO3V3_LDO | 2.9 | 3.3 | 3.6 | V |
| | MIC_BIAS | 1.8 | 2.8 | 2.8 | V |
| | IO1V8_LDO | 1.65 | 1.8 | 2.0 | V |
| | SYS_BFB | 1.35 | 1.4 | 1.5 | V |
| | DVDD_LDO | 1.15 | 1.2 | 1.25 | V |

| Operating Condition | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| | BT_IOLDO | 2.9 | 3.3 | 3.6 | V |
| | BT_BFB | 1.2 | 1.25 | 1.4 | V |
| Supply Voltage | MCU_VCC | 2.9 | 3.3 | 3.6 | V |
| | BT_VCC | 2.9 | 3.3 | 3.6 | V |

## 4.3 Power Consumption

Table 4-3 Power Consumption

| Operation Mode | Average | Max. | Unit |
|---|---|---|---|
| TX peek current (0dB) | - | 18.2 | mA |
| RX peek current | - | 9.8 | mA |
| Deep sleep current (include 128K retention RAM) | TBD | - | uA |
| Power off | TBD | - | μA |

## 4.4 IO Electrical Logical Characteristics

Table 4-4 IO Electrical Logical Characteristics

| Symbol | Parameter | Min. | Typ. | Max. | Unit | Test Conditions |
|---|---|---|---|---|---|---|
| VIL | Low-Level Input Voltage | -0.3 | - | 0.3* IO | V | IO=3.3V,<br>IO=1.8V |
| VIH | High-Level Input Voltage | 0.7*IO | - | IO+0.2 | V | IO=3.3V,<br>IO=1.8V |
| VOL | Low-Level Output Voltage | - | - | 0.3 | V | IO=3.3V,<br>IO=1.8V |
| VOH | High-Level Output Voltage | 0.7*IO | - | - | V | IO=3.3V,<br>IO=1.8V |

## 4.5 IO Resistance Parameters

table 4-5 IO Resistance Parameters

| Name | Pull-up Resistor | Pull-down Resistor | Driving Capability |
|---|---|---|---|
| PORT_PMU(ADC) | 16K | 16K | 16mA |
| GPIO (1.8V) | 12.5K | 10K | Configurable:4 /8 / 12 /16mA |
| GPIO (3.3V) | 10K | 10K | Configurable:4 /8 / 12 /16mA |

## 4.6 Crystal Oscillator

Table 4-6 Crystal Oscillator

| Parameter | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|
| Clock Frequency | 24 | 24 | 24 | MHz |
| CL Load capacitance | - | 9 | 12 | pF |
| Tolerance | - | +/-10 | - | PPM |
| Motional resistance | - | - | 60 | Ω |
| Shunt capacitance | - | - | 2 | pF |

## 4.7 Inductance

Table 4-7 Inductance

| BUCK | Inductance Value(μH) | Saturated Current(mA) | Self-Resonant Frequency (MH) | Direct Current Resistance(ohm) |
|---|---|---|---|---|
| BT_BUCK | 10 | >=80 | >=10 | <=1 |
| SYS_BUCK | 2.2 | >=300 | >=10 | <=1 |

## 4.8 ESD

table 4-8 ESD

| Mode | Value | Unit |
|---|---|---|
| HBM | ±2000 | V |
| CDM | ±500 | V |

# Abbreviations

| Abbreviations | Descriptions |
|---|---|
| AGC | Automatic Gain Control |
| ADC | Analog-to-Digital-Converter |
| GPIO | General Purpose Input Output |
| PMU | Power Management Unit |
| OSC | Oscillator |

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# Revision History

**Feedback:**

Freqchip welcomes feedback on this product and this document. If you have comments or suggestions, please send an email to docs@freqchip.com.

| Reversion Number | Reversion Date | Description |
|---|---|---|
| V0.3.0 | 2024.10.10 | Initial Draft |
| V0.4.2 | 2025.02.21 | Add IO IO Resistance Parameters<br>Modify CAN FD information |
| V0.4.3 | 2025.03.05 | Add FR3068E-D / FR3066E-D |
| V0.4.3 | 2025.05.28 | Modify FR3068E-C Pins |
| V0.4.4 | 2025.05.26 | Modify FR3068E-C pins |
| V0.4.5 | 2025.06.10 | Add FR3066EQ-D |
| V0.4.6 | 2025.06.30 | Add FR3066D-C |
| V0.4.7 | 2025.07.15 | Add FR3068EP-C |
| V0.4.8 | 2025.07.28 | Change Series FR306x-C to Series FR306x |
| V0.4.9 | 2025.09.16 | Add ESD information |
| V0.5.0 | 2026.01.21 | Update ordering information quantity |
| V0.5.1 | 2026.04.07 | Add FR3068EQ-D |
| V0.5.2 | 2026.05.20 | Add Chapter 1.2 Hardware Resources |
| V0.5.3 | 2026.07.09 | Update transport quantity |
| V0.5.4 | 2026.09.10 | Add FR3066E-D, delete FR3066EQ-D |